

Output tables for 1xN statistical comparisons.

March 10, 2025

1 Average rankings of Friedman test

Average ranks obtained by each method in the Friedman test.

Algorithm	Ranking
Random	7
NN	5
Sweep	6
SaveSeq	3
SaveParall	1.375
MJ	1.625
CMT	4

Table 1: Average Rankings of the algorithms (Friedman)

Friedman statistic (distributed according to chi-square with 6 degrees of freedom): 47.196429.
P-value computed by Friedman Test: 0.

2 Post hoc comparison (Friedman)

P-values obtained in by applying post hoc methods over the results of Friedman procedure.

i	algorithm	$z = (R_0 - R_i)/SE$	p	Holm	Finner	Li
6	Random	5.207738	0	0.008333	0.008512	0.009634
5	Sweep	4.281918	0.000019	0.01	0.016952	0.009634
4	NN	3.356098	0.000791	0.0125	0.025321	0.009634
3	CMT	2.430278	0.015087	0.016667	0.033617	0.009634
2	SaveSeq	1.504458	0.132464	0.025	0.041844	0.009634
1	MJ	0.231455	0.816961	0.05	0.05	0.05

Table 2: Post Hoc comparison Table for $\alpha = 0.05$ (FRIEDMAN)

Holm's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.025 .
Finner's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.041844 .
Li's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.009634 .

3 Adjusted P-Values (Friedman)

Adjusted P-values obtained through the application of the post hoc methods (Friedman).

i	algorithm	unadjusted p	p_{Holm}
1	Random	0	0.000001
2	Sweep	0.000019	0.000093
3	NN	0.000791	0.003162
4	CMT	0.015087	0.045262
5	SaveSeq	0.132464	0.264927
6	MJ	0.816961	0.816961

Table 3: Adjusted p -values (FRIEDMAN) (I)

i	algorithm	unadjusted p	p_{Finner}	p_{Li}
1	Random	0	0.000001	0.000001
2	Sweep	0.000019	0.000056	0.000101
3	NN	0.000791	0.00158	0.0043
4	CMT	0.015087	0.022545	0.07615
5	SaveSeq	0.132464	0.156772	0.41985
6	MJ	0.816961	0.816961	0.816961

Table 4: Adjusted p -values (FRIEDMAN) (II)